

Geometry-Informed Predictive Cost Adaptive Control with Online Full-Inertia Identification

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Adaptive rigid-body attitude control must accommodate nonlinear rotational dynamics, uncertain mass properties, and actuator constraints. This paper develops and evaluates a geometry-informed realization of predictive cost adaptive control (PCAC) that identifies all six independent entries of a non-principal-axis symmetric inertia matrix online. An interval-integrated inverse-dynamics regression and variable-rate-forgetting recursive least squares avoid numerical differentiation of measured angular velocity, while bounded positive-definite projection maintains a physically admissible estimate. The projected estimate is embedded in a nonlinear rigid-body sampled predictor that propagates attitude multiplicatively with the quaternion exponential and is expressed in local principal-rotation coordinates. Freezing the predictor Jacobian over the horizon yields a convex quadratic program subject to torque and torque-increment constraints. In 200 large-angle Monte Carlo trials, the method achieves 100% stabilization success and a 0.09% median final inertia error. Following an abrupt inertia change, the final inertia error reaches 0.04%, and the final tracking errors approach those of an exact-inertia benchmark. Under continuous inertia variation, tracking remains close to the benchmark, whereas the estimator retains a 7.43% mean inertia error with visible lag and bias. These results indicate that variable-rate forgetting supports abrupt-change readaptation but activates weakly under slow variation and does not replace sufficient excitation.

Nomenclature

$\mathbf{A}_k, \mathbf{B}_k, \mathbf{c}_k$	=	state matrix, input matrix, and affine term of the predictor; mixed units
$\mathbf{b}_{J,k}$	=	sampled torque impulse, $\mathbf{b}_{J,k} = T_s \boldsymbol{\tau}_{k-1}$, N m s
C_k	=	local quaternion-state coordinate map at sampling instant k ; units not applicable
$\mathbf{e}_R$	=	principal rotation-vector attitude error, rad
$\mathbf{e}_\omega$	=	body-frame angular-velocity tracking error, rad s ⁻¹
$\hat{\mathbf{F}}_{s,k}$	=	adaptive nonlinear sampled state-transition map; mixed units

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$\widehat{\mathbf{F}}_{\zeta,k}$	= adaptive nonlinear sampled predictor in local coordinates; mixed units
$\mathbf{G}(\mathbf{q})$	= quaternion kinematic matrix; dimensionless
h	= internal integration-step duration, $h = T_s/N_s$, s
$\mathbf{J}$	= full symmetric positive-definite inertia matrix, kg m ²
$\widehat{\mathbf{J}}_k$	= projected inertia estimate used for prediction and control, kg m ²
$\underline{J}, \overline{J}$	= lower and upper admissible inertia eigenvalue bounds, kg m ²
$\mathcal{J}$	= admissible set of bounded positive-definite inertia matrices; elements have units kg m ²
N	= prediction-horizon length; dimensionless
N_f	= residual-window length used for variable-rate forgetting; dimensionless
N_s	= number of internal integration steps per sampling interval; dimensionless
$\mathbf{P}_k$	= recursive least-squares covariance matrix, s ²
$\mathbf{q}, \mathbf{q}_d, \mathbf{q}_e$	= actual, desired, and multiplicative-error unit quaternions; dimensionless
$\mathbf{Q}_i, \mathbf{R}_\tau$	= state-tracking and torque-increment weighting matrices; units consistent with the cost
$\mathbf{R}(\mathbf{q})$	= rotation matrix mapping body-frame components to inertial-frame components; dimensionless
T_s	= controller sampling period, s
$\mathbf{x}$	= attitude state, $\mathbf{x} = [\mathbf{q}^\top \boldsymbol{\omega}^\top]^\top$; mixed units
$\mathbf{Y}_J$	= full-inertia inverse-dynamics regressor, s ⁻²
$\boldsymbol{\alpha}$	= body-frame angular acceleration, $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}$, rad s ⁻²
β_k	= variable covariance-inflation rate; β_k^{-1} is the instantaneous forgetting factor; dimensionless
$\boldsymbol{\varepsilon}_{J,k}$	= a priori inertia-regression error, N m s
$\boldsymbol{\Phi}_{J,k}$	= sampled full-inertia regressor, s ⁻¹
$\boldsymbol{\theta}_J$	= six-dimensional full-inertia parameter vector, kg m ²
$\widehat{\boldsymbol{\theta}}_{J,k}^{\text{rls}}$	= unconstrained recursive least-squares inertia estimate, kg m ²
$\boldsymbol{\tau}$	= commanded body torque about the center of mass, N m
$\Delta\boldsymbol{\tau}$	= commanded torque increment, N m
$\boldsymbol{\vartheta}$	= local principal rotation vector relative to $\mathbf{q}_k$, rad
$\boldsymbol{\omega}, \boldsymbol{\omega}_d$	= actual and desired body angular velocities, rad s ⁻¹
$\boldsymbol{\omega}_{\text{loc}}$	= angular velocity expressed in the fixed local body frame, rad s ⁻¹
$\boldsymbol{\zeta}$	= local attitude state, $\boldsymbol{\zeta} = [\boldsymbol{\vartheta}^\top \boldsymbol{\omega}_{\text{loc}}^\top]^\top$; mixed units
Σ_i, Σ_b	= inertial and body-fixed reference frames, respectively; units not applicable

$\mathbf{e}_j$	= j th Euclidean unit vector; dimensionless
$\mathcal{E}_{J,k}$	= normalized root-mean-square inertia-regression residual; dimensionless
$\mathbf{f}, \mathbf{f}_\omega$	= parameterized attitude and angular-velocity vector fields, respectively; mixed units and rad s^{-2}
$\hat{\mathbf{f}}_{\omega,k}$	= estimated angular-velocity vector field, rad s^{-2}
$\mathbf{I}_n, \mathbf{0}_n, \mathbf{1}_n$	= n -dimensional identity matrix, zero vector, and vector of ones, respectively; dimensionless
$\mathbf{J}_A, \mathbf{J}_B$	= endpoint inertia matrices used in the numerical evaluation, kg m^2
$\hat{\mathbf{J}}_k^{\text{rls}}$	= unconstrained inertia-matrix estimate, kg m^2
$j_1(t), j_2(t), j_3(t)$	= time-varying principal inertias in numerical Case 3, kg m^2
$\mathbf{L}_k$	= inflated covariance matrix, $\mathbf{L}_k = \beta_k \mathbf{P}_{k-1}$, s^2
$\mathcal{L}$	= principal quaternion-to-rotation-vector map; output in rad
n_k, n_θ	= active residual-window size and number of estimated inertia parameters; dimensionless
$p_{P,0}$	= initial covariance scale, $\mathbf{P}_0 = p_{P,0} \mathbf{I}_6$, s^2
$\mathbf{p}, p_0, \mathbf{p}_v$	= generic scalar-first quaternion and its scalar and vector parts; dimensionless
$q_0, \mathbf{q}_v$	= scalar and vector parts of a generic quaternion; dimensionless
$\bar{\mathbf{q}}_e, \bar{q}_{e,0}, \bar{\mathbf{q}}_{e,v}$	= principal quaternion-error representative and its scalar and vector parts; dimensionless
$q_{e,0}, \mathbf{q}_{e,v}$	= scalar and vector parts of the multiplicative quaternion error; dimensionless
$\mathbf{R}_k, \mathbf{R}_e$	= current attitude rotation matrix and multiplicative rotation-error matrix; dimensionless
r_s	= soft angular-velocity constraint penalty; units consistent with the cost
s_e	= sign selecting the principal quaternion-error representative; dimensionless
$\mathbf{s}_{\omega,i k}$	= nonnegative angular-velocity constraint slack vector, rad s^{-1}
t, t_k	= continuous time and sampling time $t_k = kT_s$, s
$\mathbf{U}_A, \mathbf{V}_k$	= Case 3 principal-axis matrix and inertia-estimate eigenvector matrix, respectively; dimensionless
$\mathbf{x}^+, \mathbf{x}_{d,i k}$	= sampled successor state and desired predicted state; mixed units
$\mathbf{Y}_\alpha, \mathbf{Y}_\omega$	= angular-acceleration and angular-velocity components of $\mathbf{Y}_J$, s^{-2}
$\delta_{\zeta,j}, \delta_{\tau,j}$	= finite-difference perturbations for local-state and torque components; mixed units and N m , respectively
η, γ	= variable-rate-forgetting sensitivity and saturation parameters; dimensionless
ℓ	= internal integration-step index; dimensionless
$\mu_{j,k}, \bar{\mu}_{j,k}$	= unconstrained and clipped inertia eigenvalues, kg m^2

$\mathbf{v}_{J,k}$	=	sampled regression disturbance containing quadrature, measurement, actuator, and model-mismatch effects, N m s
$\Phi_{J,k}^*$	=	exact integrated full-inertia regressor, s ⁻¹
ρ, ξ	=	norm and argument of the quaternion exponential map, rad
$\sigma_{\mathcal{E}}$	=	reference impulse-residual magnitude, N m s
$\hat{\boldsymbol{\theta}}_{J,k}$	=	projected inertia-parameter estimate, kg m ²
$\boldsymbol{\tau}_{\min}, \boldsymbol{\tau}_{\max}$	=	componentwise commanded-torque limits, N m
$\Delta\boldsymbol{\tau}_{\min}, \Delta\boldsymbol{\tau}_{\max}$	=	componentwise commanded-torque-increment limits, N m
$\omega_{\max}$	=	componentwise soft angular-velocity limit, rad s ⁻¹
$\zeta_{d,i k}$	=	desired local attitude state at prediction step i ; mixed units
$\mathbb{R}^n, \mathbb{S}^3, \text{SO}(3)$	=	n -dimensional Euclidean space, unit-quaternion sphere, and three-dimensional special orthogonal group; units not applicable

Subscripts and Operators

k	=	sampling-time index
$i k$	=	prediction step i computed at sampling instant k
d, e, loc, rls, s	=	desired, tracking-error, local-coordinate, unconstrained-estimate, and sampled-map quantities, respectively
$(\cdot)^\times$	=	skew-symmetric matrix satisfying $\mathbf{a}^\times \mathbf{b} = \mathbf{a} \times \mathbf{b}$
$(\cdot)^*, (\cdot)^\star$	=	quaternion conjugate and optimal quantity, respectively
$(\cdot)^\top, (\cdot)^+$	=	transpose and sampled successor, respectively
$\ \cdot\ _2, \ \mathbf{v}\ _{\mathbf{M}}$	=	Euclidean and weighted norms, respectively, with $\ \mathbf{v}\ _{\mathbf{M}}^2 = \mathbf{v}^\top \mathbf{M} \mathbf{v}$
$\kappa_2(\cdot)$	=	matrix condition number induced by the Euclidean norm; dimensionless
$\otimes$	=	Hamilton quaternion product
$\text{Exp}_q(\cdot), \text{sinc}(\cdot)$	=	quaternion exponential map and unnormalized sinc function
$\text{proj}_{\mathcal{J}}(\cdot), \text{sat}_{\gamma}(\cdot)$	=	eigenvalue-clipping projection and scalar saturation operators

I. Introduction

RIGID-BODY attitude control combines nonlinear rotational dynamics with a non-Euclidean configuration space. Large-angle maneuvers require attitude representations and numerical propagation that respect the geometry of SO(3). Uncertain mass properties can alter gyroscopic coupling and control effectiveness, while practical actuators impose torque-magnitude and torque-increment limits. Model predictive control (MPC) handles such constraints directly through receding-horizon optimization, but its performance depends on prediction-model fidelity [1, 2].

Adaptive attitude control has addressed inertia uncertainty through several complementary approaches. Adaptive asymptotic tracking has been combined with online identification of the full spacecraft inertia matrix [3], and quaternion-based adaptive tracking has been developed for unknown inertia without angular-velocity measurements [4]. Inertia-free tracking with disturbance rejection and almost-global stabilization [5] and robust adaptive tracking with inertia adaptation [6] have both been formulated directly on $SO(3)$. Related methods address uncertain time-varying inertia [7], adaptive optimal control with inertia-matrix identification [8], and recursive full-inertia estimation subject to positive-definiteness and physical constraints [9]. Quaternion-based adaptive output-feedback control has also incorporated uncertain inertia and actuator-magnitude constraints for flexible spacecraft [10]. These lines establish several forms of adaptive attitude control, geometric adaptation, and physical inertia estimation. They also distinguish tracking from parameter convergence: accurate tracking does not guarantee accurate inertia recovery, which requires sufficiently informative motion.

Manifold-aware control provides a complementary treatment of attitude geometry. Geometric tracking control for rigid-body aerial vehicles has been posed directly on $SE(3)$ [11]. Constrained spacecraft attitude control has been formulated on $SO(3)$ using nonlinear MPC and a Lie-group variational model [12]. MPC has also been developed for systems on smooth manifolds [2] and for an underactuated spacecraft with two reaction wheels [13]. Collectively, these methods provide well-developed geometric and constraint-aware control formulations, primarily using a specified predictive model.

Adaptive MPC and online model refinement span a broader class of predictive control methods. Set-membership identification has been combined with constraint-robust receding-horizon control for uncertain multi-input multi-output (MIMO) systems [14]. Robust MPC formulations recursively update model uncertainty and point estimates while preserving stability and constraint satisfaction [15]. Other adaptive MPC methods combine recursive least squares (RLS) with tube-based constrained prediction [16]. Persistent-excitation constraints have also been introduced to reconcile predictive regulation with online parameter learning [17]. Within this broader adaptive-MPC landscape, indirect architectures can continually update the model supplied to receding-horizon optimization.

Predictive cost adaptive control (PCAC) is the specific indirect adaptive architecture underlying the present work. Its foundational development examined persistency, consistency, and exigency in closed-loop identification [18]. PCAC continually identifies a predictive model from reference commands, disturbances, and corrective control, and supplies that model to constrained predictive optimization. The standard formulation does not require a separately designed probing signal. Aerospace applications include active flutter suppression for the NASA benchmark flutter model [19] and active flow control with continual closed-loop identification [20]. Experimental quadcopter flight tests have implemented PCAC with online identification and constrained predictive optimization [21]. These applications use continual model adaptation when a control-oriented model is uncertain or may change during operation.

Spacecraft and nonlinear applications provide the closest PCAC parallels. Fully actuated and underactuated

detumbling under unknown inertia and sensor–actuator alignment identifies a generic MIMO linear input-output model rather than the physical inertia matrix [22]. Spacecraft attitude PCAC without prior modeling likewise minimizes the required model information across several attitude representations [23]. Nonlinear PCAC learns pseudo-linear input-output representations using polynomial, Fourier, and cubic-spline observables, and couples them with iterative nonlinear MPC [24]. The present formulation makes a complementary modeling choice: it retains rigid-body mechanics and quaternion kinematics, identifies only the six physical inertia parameters, and freezes a local Jacobian so that the online horizon problem remains a convex quadratic program (QP).

Structure-informed PCAC provides direct methodological continuity. A sparse quadrotor formulation uses the known pattern of a hover-linearized model to reduce the number of coefficients identified by RLS, although it employs Euler-angle coordinates and a principal-axis, diagonal-inertia model [25]. A subsequent formulation constructs nonlinear sampled maps from structured Taylor dictionaries and forms Jacobian-frozen affine predictors for QP-based MPC [26]. The present development continues this progression from sparse structure-informed identification, through nonlinear sampled prediction and Jacobian freezing, to physical full-inertia adaptation.

Existing work thus provides adaptive attitude control under unknown inertia, geometric and constrained attitude control, adaptive MPC with online model refinement, generic PCAC under high model uncertainty, and recursive full-inertia estimation. The question considered here is how physical full-inertia identification can be incorporated into a geometry-informed predictive architecture while retaining the QP structure of PCAC. This paper addresses that question through three contributions. First, an integrated inverse-dynamics regression avoids numerical differentiation of measured angular velocity, and variable-rate-forgetting recursive least squares (VRF-RLS) identifies all six independent entries of a non-principal-axis symmetric inertia matrix. A bounded positive-definite projection supplies a nonsingular estimate for prediction and control. Second, the projected estimate is inserted into the nonlinear rigid-body forward dynamics, while known quaternion kinematics are retained. Attitude is propagated multiplicatively with the quaternion exponential, and the sampled nonlinear map is expressed in local principal-rotation coordinates. Third, the map Jacobian is evaluated at the current operating point and frozen over the prediction horizon. The resulting affine predictor preserves convex quadratic optimization with componentwise torque and torque-increment constraints. Thus, inverse dynamics support identification, nonlinear forward dynamics support prediction, and local affine dynamics support optimization. Complete parameter convergence is not asserted without sufficient excitation. Numerical simulations examine large-angle stabilization, an abrupt inertia change, and continuously varying inertia while distinguishing tracking performance from parameter-estimation accuracy.

II. Problem Formulation

Consider the rotational motion of a rigid body. Let Σ_i denote an inertial frame and let Σ_b denote a body-fixed frame whose origin is located at the center of mass. Let $\mathbf{R} \in \text{SO}(3)$ map body-frame components to inertial-frame

90 components. Thus, $\mathbf{R}$ is used as a passive component transformation, so that, for any $\mathbf{a} \in \mathbb{R}^3$,

$$\mathbf{a}^i = \mathbf{R} \mathbf{a}^b. \quad (1)$$

91 Let $\boldsymbol{\omega} \in \mathbb{R}^3$ denote the angular velocity of Σ_b relative to Σ_i , expressed in Σ_b .

92 For $\mathbf{a} = [a_1 \ a_2 \ a_3]^\top \in \mathbb{R}^3$, define

$$\mathbf{a}^\times := \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}, \quad (2)$$

93 so that $\mathbf{a}^\times \mathbf{b} = \mathbf{a} \times \mathbf{b}$ for all $\mathbf{b} \in \mathbb{R}^3$.

94 A. Quaternion Algebra

95 The attitude is represented by the scalar-first unit quaternion

$$\mathbf{q} := \begin{bmatrix} q_0 \\ \mathbf{q}_v \end{bmatrix} \in \mathbb{S}^3, \quad \mathbb{S}^3 := \{ \mathbf{q} \in \mathbb{R}^4 : \mathbf{q}^\top \mathbf{q} = 1 \}, \quad (3)$$

96 where $q_0 \in \mathbb{R}$ and $\mathbf{q}_v \in \mathbb{R}^3$. The quaternion $\mathbf{q}$ represents the attitude of Σ_b relative to Σ_i .

97 For

$$\mathbf{p} = \begin{bmatrix} p_0 \\ \mathbf{p}_v \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} q_0 \\ \mathbf{q}_v \end{bmatrix}, \quad (4)$$

98 the Hamilton product is

$$\mathbf{p} \otimes \mathbf{q} = \begin{bmatrix} p_0 q_0 - \mathbf{p}_v^\top \mathbf{q}_v \\ p_0 \mathbf{q}_v + q_0 \mathbf{p}_v + \mathbf{p}_v^\times \mathbf{q}_v \end{bmatrix}. \quad (5)$$

99 The quaternion conjugate is

$$\mathbf{q}^* := \begin{bmatrix} q_0 \\ -\mathbf{q}_v \end{bmatrix}. \quad (6)$$

100 For $\mathbf{q} \in \mathbb{S}^3$, $\mathbf{q}^{-1} = \mathbf{q}^*$.

101 The rotation matrix associated with $\mathbf{q}$ is

$$\mathbf{R}(\mathbf{q}) = (q_0^2 - \mathbf{q}_v^\top \mathbf{q}_v) \mathbf{I}_3 + 2\mathbf{q}_v \mathbf{q}_v^\top + 2q_0 \mathbf{q}_v^\times. \quad (7)$$

102 The map from $\mathbb{S}^3$ to $\text{SO}(3)$ is two-to-one:

$$\mathbf{R}(\mathbf{q}) = \mathbf{R}(-\mathbf{q}). \quad (8)$$

103 Hence, $\mathbf{q}$ and $-\mathbf{q}$ represent the same physical attitude.

104 B. Rigid-Body Attitude Dynamics

105 With the conventions above, the quaternion kinematics are

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} = \frac{1}{2} \mathbf{G}(\mathbf{q}) \boldsymbol{\omega}, \quad (9)$$

106 where

$$\mathbf{G}(\mathbf{q}) := \begin{bmatrix} -\mathbf{q}_v^\top \\ q_0 \mathbf{I}_3 + \mathbf{q}_v^\times \end{bmatrix}. \quad (10)$$

107 Equation (9) is equivalent to

$$\dot{\mathbf{R}} = \mathbf{R} \boldsymbol{\omega}^\times. \quad (11)$$

108 The matrix $\mathbf{G}(\mathbf{q})$ satisfies

$$\mathbf{q}^\top \mathbf{G}(\mathbf{q}) = \mathbf{0}_{1 \times 3}. \quad (12)$$

109 Therefore,

$$\frac{d}{dt} (\mathbf{q}^\top \mathbf{q}) = 2 \mathbf{q}^\top \dot{\mathbf{q}} = 0. \quad (13)$$

110 The continuous-time quaternion dynamics preserve $\mathbf{q} \in \mathbb{S}^3$.

111 Let $\boldsymbol{\tau} \in \mathbb{R}^3$ denote the commanded torque about the center of mass, expressed in Σ_b . The rotational dynamics are

$$\mathbf{J} \dot{\boldsymbol{\omega}} = -\boldsymbol{\omega}^\times \mathbf{J} \boldsymbol{\omega} + \boldsymbol{\tau}. \quad (14)$$

112 The body axes are not assumed to coincide with the principal axes. The inertia matrix is

$$\mathbf{J} = \begin{bmatrix} J_{11} & J_{12} & J_{13} \\ J_{12} & J_{22} & J_{23} \\ J_{13} & J_{23} & J_{33} \end{bmatrix}, \quad \mathbf{J} = \mathbf{J}^\top \succ \mathbf{0}. \quad (15)$$

113 The six independent entries of $\mathbf{J}$ are unknown. The identification and prediction models assume that the inertia is

constant over each sampling interval and fixed over each prediction horizon. Thus, constant and piecewise-constant inertia profiles are represented directly. Numerical Case 3 additionally evaluates continuous inertia variation, for which the inertia-rate contribution is treated as model mismatch in the sampled identification regression.

The attitude state is

$$\mathbf{x} := \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\omega} \end{bmatrix} \in \mathbb{S}^3 \times \mathbb{R}^3, \quad (16)$$

and the control input is $\boldsymbol{\tau}$.

C. Tracking Objective and Constraints

Let $\mathbf{q}_d(t) \in \mathbb{S}^3$ denote the desired attitude, and let $\boldsymbol{\omega}_d(t) \in \mathbb{R}^3$ denote the desired angular velocity expressed in the desired body frame. The reference is assumed continuously differentiable and kinematically consistent:

$$\dot{\mathbf{q}}_d = \frac{1}{2} \mathbf{q}_d \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega}_d \end{bmatrix}. \quad (17)$$

Define the multiplicative quaternion error

$$\mathbf{q}_e := \mathbf{q}_d^* \otimes \mathbf{q} = \begin{bmatrix} q_{e,0} \\ \mathbf{q}_{e,v} \end{bmatrix}. \quad (18)$$

The corresponding rotation error is

$$\mathbf{R}_e := \mathbf{R}(\mathbf{q}_e) = \mathbf{R}(\mathbf{q}_d)^\top \mathbf{R}(\mathbf{q}). \quad (19)$$

The matrix $\mathbf{R}_e$ maps current-body components to desired-body components.

To select the quaternion representative associated with the principal rotation, define

$$s_e := \begin{cases} 1, & q_{e,0} \geq 0, \\ -1, & q_{e,0} < 0, \end{cases} \quad \bar{\mathbf{q}}_e := s_e \mathbf{q}_e = \begin{bmatrix} \bar{q}_{e,0} \\ \bar{\mathbf{q}}_{e,v} \end{bmatrix}. \quad (20)$$

The attitude-error vector is

$$\mathbf{e}_R := \begin{cases} \mathbf{0}_3, & \|\bar{\mathbf{q}}_{e,v}\| = 0, \\ 2 \operatorname{atan2}(\|\bar{\mathbf{q}}_{e,v}\|, \bar{q}_{e,0}) \frac{\bar{\mathbf{q}}_{e,v}}{\|\bar{\mathbf{q}}_{e,v}\|}, & \|\bar{\mathbf{q}}_{e,v}\| > 0. \end{cases} \quad (21)$$

127 Its norm is the principal rotation angle between the desired and actual attitudes:

$$\|\mathbf{e}_R\| \in [0, \pi]. \quad (22)$$

128 At an attitude error of exactly π rad, the rotation axis is not unique. The local predictive model developed below is
129 therefore constructed on $\|\mathbf{e}_R\| < \pi$.

130 The angular-velocity error is

$$\mathbf{e}_\omega := \boldsymbol{\omega} - \mathbf{R}_e^\top \boldsymbol{\omega}_d. \quad (23)$$

131 The factor $\mathbf{R}_e^\top$ expresses $\boldsymbol{\omega}_d$ in the current body frame. With this definition, the quaternion-error kinematics satisfy

$$\dot{\mathbf{q}}_e = \frac{1}{2} \mathbf{q}_e \otimes \begin{bmatrix} 0 \\ \mathbf{e}_\omega \end{bmatrix}. \quad (24)$$

132 Let $T_s > 0$ denote the sampling period, and let $t_k = kT_s$. The quaternion and angular velocity are assumed available
133 at each sampling instant. The commanded torque is held constant over each sampling interval:

$$\boldsymbol{\tau}(t) = \boldsymbol{\tau}_k, \quad t \in [t_k, t_{k+1}). \quad (25)$$

134 Define

$$\Delta \boldsymbol{\tau}_k := \boldsymbol{\tau}_k - \boldsymbol{\tau}_{k-1}. \quad (26)$$

135 The torque and torque increment satisfy the componentwise constraints

$$\boldsymbol{\tau}_{\min} \leq \boldsymbol{\tau}_k \leq \boldsymbol{\tau}_{\max}, \quad (27)$$

136 and

$$\Delta \boldsymbol{\tau}_{\min} \leq \Delta \boldsymbol{\tau}_k \leq \Delta \boldsymbol{\tau}_{\max}. \quad (28)$$

137 The control objective is to achieve

$$\mathbf{e}_{R,k} \rightarrow \mathbf{0}_3, \quad \mathbf{e}_{\omega,k} \rightarrow \mathbf{0}_3, \quad (29)$$

138 while satisfying (27) and (28). The inertia matrix is estimated online. The known quaternion kinematics are retained
139 explicitly rather than identified.

III. Inertia Parameterization

The quaternion kinematics in Eq. (9) are known and contain no uncertain coefficients. The model uncertainty is confined to the inertia matrix $\mathbf{J}$. This section derives an exact linear parameterization of the full rigid-body inertia. No diagonal-inertia or principal-axis assumption is introduced.

A. Full-Inertia Parameter Vector

Define the inertia parameter vector

$$\boldsymbol{\theta}_J := \begin{bmatrix} J_{11} & J_{22} & J_{33} & J_{12} & J_{13} & J_{23} \end{bmatrix}^\top \in \mathbb{R}^6. \quad (30)$$

The corresponding inertia matrix is

$$\mathbf{J}(\boldsymbol{\theta}_J) := \begin{bmatrix} J_{11} & J_{12} & J_{13} \\ J_{12} & J_{22} & J_{23} \\ J_{13} & J_{23} & J_{33} \end{bmatrix}. \quad (31)$$

This parameterization enforces symmetry by construction. Positive definiteness is not guaranteed by an unconstrained parameter estimate and will be enforced by projection during online identification.

B. Linear Inverse-Dynamics Regression

Let

$$\boldsymbol{\alpha} := \dot{\boldsymbol{\omega}} = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 \end{bmatrix}^\top \quad (32)$$

denote the angular acceleration expressed in the body-fixed frame. Rearranging Eq. (14) gives

$$\boldsymbol{\tau} = \mathbf{J}(\boldsymbol{\theta}_J)\boldsymbol{\alpha} + \boldsymbol{\omega}^\times \mathbf{J}(\boldsymbol{\theta}_J)\boldsymbol{\omega}. \quad (33)$$

Collecting the six independent inertia entries yields

$$\boldsymbol{\tau} = \mathbf{Y}_J(\boldsymbol{\omega}, \boldsymbol{\alpha})\boldsymbol{\theta}_J, \quad (34)$$

where

$$\mathbf{Y}_J(\boldsymbol{\omega}, \boldsymbol{\alpha}) := \begin{bmatrix} \alpha_1 & -\omega_2\omega_3 & \omega_2\omega_3 & \alpha_2 - \omega_1\omega_3 & \alpha_3 + \omega_1\omega_2 & \omega_2^2 - \omega_3^2 \\ \omega_1\omega_3 & \alpha_2 & -\omega_1\omega_3 & \alpha_1 + \omega_2\omega_3 & \omega_3^2 - \omega_1^2 & \alpha_3 - \omega_1\omega_2 \\ -\omega_1\omega_2 & \omega_1\omega_2 & \alpha_3 & \omega_1^2 - \omega_2^2 & \alpha_1 - \omega_2\omega_3 & \alpha_2 + \omega_1\omega_3 \end{bmatrix}. \quad (35)$$

The regressor $\mathbf{Y}_J \in \mathbb{R}^{3 \times 6}$ is nonlinear in the measured angular motion but independent of the unknown inertia parameters. Equation (34) is therefore linear in θ_J and is suitable for recursive least-squares (RLS) identification. The parameterization is exact and does not require a Taylor approximation.

Each measurement instant provides three equations for six unknown parameters. Complete inertia identification therefore relies on data collected over multiple sampling intervals and requires sufficiently rich rotational motion.

C. Identification and Prediction Models

Equation (34) is an inverse-dynamics model. Its direct implementation would require $\alpha = \dot{\omega}$, which may be sensitive to measurement noise. The online identification method developed in the next section integrates the rigid-body dynamics over each sampling interval. This removes the need to differentiate the measured angular velocity while preserving linearity in θ_J .

The forward angular-velocity dynamics are

$$\dot{\omega} = \mathbf{f}_\omega(\omega, \tau; \theta_J) := \mathbf{J}^{-1}(\theta_J) [-\omega^\times \mathbf{J}(\theta_J) \omega + \tau]. \quad (36)$$

Unlike the inverse-dynamics regression, the forward model is not linear in θ_J because it contains $\mathbf{J}^{-1}(\theta_J)$.

The inverse model is therefore used for parameter identification. The resulting estimate is then used to reconstruct the inertia matrix and evaluate the nonlinear forward model. Combining Eqs. (9) and (36) gives

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \tau; \theta_J) := \begin{bmatrix} \frac{1}{2} \mathbf{G}(\mathbf{q}) \omega \\ \mathbf{f}_\omega(\omega, \tau; \theta_J) \end{bmatrix}. \quad (37)$$

Only the angular-dynamics block depends on the unknown parameters. The quaternion kinematics retain their known Hamilton-product structure and are not identified.

IV. Online Full-Inertia Identification

The inertia parameters are identified from sampled angular-velocity and torque measurements. Direct use of Eq. (34) would require the angular acceleration. Instead, the inverse dynamics are integrated over each sampling interval. The resulting regression remains linear in the six inertia parameters and avoids numerical differentiation.

A. Sampled Inverse-Dynamics Regression

Separate the angular-acceleration and angular-velocity terms in the continuous-time regressor:

$$\mathbf{Y}_J(\omega, \alpha) = \mathbf{Y}_\alpha(\alpha) + \mathbf{Y}_\omega(\omega), \quad (38)$$

176 where

$$\mathbf{Y}_\alpha(\alpha) := \mathbf{Y}_J(\mathbf{0}_3, \alpha), \quad \mathbf{Y}_\omega(\omega) := \mathbf{Y}_J(\omega, \mathbf{0}_3). \quad (39)$$

177 The map $\mathbf{Y}_\alpha$ is linear in α , whereas $\mathbf{Y}_\omega$ contains the quadratic angular-velocity terms.

178 Consider the sampling interval $[t_{k-1}, t_k]$, where $t_k = kT_s$. The inertia matrix is assumed constant over this interval.

179 Integrating Eq. (34) gives

$$\int_{t_{k-1}}^{t_k} \boldsymbol{\tau}(t) dt = \boldsymbol{\Phi}_{J,k}^* \boldsymbol{\theta}_J, \quad (40)$$

180 where

$$\boldsymbol{\Phi}_{J,k}^* := \mathbf{Y}_\alpha(\omega_k - \omega_{k-1}) + \int_{t_{k-1}}^{t_k} \mathbf{Y}_\omega(\omega(t)) dt. \quad (41)$$

181 Under the zero-order-hold torque in Eq. (25), define

$$\mathbf{b}_{J,k} := T_s \boldsymbol{\tau}_{k-1} \in \mathbb{R}^3. \quad (42)$$

182 The angular-velocity integral is approximated using the trapezoidal rule:

$$\boldsymbol{\Phi}_{J,k} := \mathbf{Y}_\alpha(\omega_k - \omega_{k-1}) + \frac{T_s}{2} [\mathbf{Y}_\omega(\omega_{k-1}) + \mathbf{Y}_\omega(\omega_k)]. \quad (43)$$

183 The sampled regression is

$$\mathbf{b}_{J,k} = \boldsymbol{\Phi}_{J,k} \boldsymbol{\theta}_J + \boldsymbol{\nu}_{J,k}, \quad (44)$$

184 where $\boldsymbol{\Phi}_{J,k} \in \mathbb{R}^{3 \times 6}$ and $\boldsymbol{\nu}_{J,k} \in \mathbb{R}^3$ contains quadrature, measurement, and actuator errors.

185 The regression depends only on the body-frame angular velocity and applied body torque. Quaternion measurements
186 are not required for the inertia update. Higher-order quadrature can be used without changing the identification
187 algorithm.

188 **B. Variable-Rate-Forgetting Recursive Least Squares**

189 Initialize

$$\hat{\boldsymbol{\theta}}_{J,0}^{\text{rls}} \in \mathbb{R}^6, \quad \mathbf{P}_0 = p_{P,0} \mathbf{I}_6, \quad p_{P,0} > 0, \quad (45)$$

190 where $\hat{\boldsymbol{\theta}}_{J,k}^{\text{rls}}$ denotes the unconstrained RLS estimate and $\mathbf{P}_k \in \mathbb{R}^{6 \times 6}$ is the covariance matrix.

191 For each $k \geq 1$, define the a priori regression error

$$\boldsymbol{\varepsilon}_{J,k} := \mathbf{b}_{J,k} - \boldsymbol{\Phi}_{J,k} \hat{\boldsymbol{\theta}}_{J,k-1}^{\text{rls}}. \quad (46)$$

192 The variable-rate-forgetting recursive least-squares (VRF-RLS) recursion of [27] is used. Adapting the authors'
 193 residual-triggered construction to the normalized vector impulse residual, define

$$n_k := \min\{k, N_f\}, \quad (47)$$

194 where $N_f \in \mathbb{N}$ is the residual-window length. Let $\sigma_\varepsilon > 0$ denote a reference impulse-residual magnitude. The
 195 normalized root-mean-square residual is

$$\mathcal{E}_{J,k} := \left[\frac{1}{n_k} \sum_{i=k-n_k+1}^k \frac{\|\boldsymbol{\varepsilon}_{J,i}\|_2^2}{\sigma_\varepsilon^2} \right]^{1/2}. \quad (48)$$

196 For $\eta > 0$ and $\gamma > 0$, define

$$\beta_k := \begin{cases} 1 + \eta \operatorname{sat}_\gamma(\mathcal{E}_{J,k}), & \mathcal{E}_{J,k} > 1, \\ 1, & \mathcal{E}_{J,k} \leq 1, \end{cases} \quad (49)$$

197 where

$$\operatorname{sat}_\gamma(s) := \min\{s, \gamma\}, \quad s \geq 0. \quad (50)$$

198 The quantity β_k^{-1} is the instantaneous forgetting factor. Therefore,

$$\frac{1}{1 + \eta\gamma} \leq \beta_k^{-1} \leq 1. \quad (51)$$

199 For $\mathcal{E}_{J,k} \leq 1$, forgetting is inactive. A larger residual increases β_k , up to the prescribed saturation level, and places
 200 greater weight on recent data.

201 Define the inflated covariance

$$\mathbf{L}_k := \beta_k \mathbf{P}_{k-1}. \quad (52)$$

202 The vector-output VRF-RLS covariance update is

$$\mathbf{P}_k = \mathbf{L}_k - \mathbf{L}_k \boldsymbol{\Phi}_{J,k}^\top \left(\mathbf{I}_3 + \boldsymbol{\Phi}_{J,k} \mathbf{L}_k \boldsymbol{\Phi}_{J,k}^\top \right)^{-1} \boldsymbol{\Phi}_{J,k} \mathbf{L}_k. \quad (53)$$

203 The parameter update is

$$\widehat{\boldsymbol{\theta}}_{J,k}^{\text{rls}} = \widehat{\boldsymbol{\theta}}_{J,k-1}^{\text{rls}} + \mathbf{P}_k \boldsymbol{\Phi}_{J,k}^\top \boldsymbol{\varepsilon}_{J,k}. \quad (54)$$

204 The matrix inverted in Eq. (53) has dimension 3×3 . In finite precision, $\mathbf{P}_k$ is symmetrized after the update.

C. Inertia Projection and Implementation

The unconstrained RLS estimate does not necessarily define a positive-definite inertia matrix. Define the admissible set

$$\mathcal{J} := \{ \mathbf{J} = \mathbf{J}^\top : \underline{\mathbf{J}}_3 \preceq \mathbf{J} \preceq \bar{\mathbf{J}}_3 \}, \quad (55)$$

where

$$0 < \underline{J} < \bar{J}. \quad (56)$$

Reconstruct the unconstrained inertia estimate:

$$\widehat{\mathbf{J}}_k^{\text{rls}} := \mathbf{J} \left(\widehat{\boldsymbol{\theta}}_{J,k}^{\text{rls}} \right). \quad (57)$$

Let

$$\widehat{\mathbf{J}}_k^{\text{rls}} = \mathbf{V}_k \text{diag}(\mu_{1,k}, \mu_{2,k}, \mu_{3,k}) \mathbf{V}_k^\top \quad (58)$$

be its eigendecomposition. Define the clipped eigenvalues

$$\bar{\mu}_{j,k} := \min \{ \max \{ \mu_{j,k}, \underline{J} \}, \bar{J} \}, \quad j \in \{1, 2, 3\}. \quad (59)$$

The projected estimate is

$$\widehat{\mathbf{J}}_k := \mathbf{V}_k \text{diag}(\bar{\mu}_{1,k}, \bar{\mu}_{2,k}, \bar{\mu}_{3,k}) \mathbf{V}_k^\top \in \mathcal{J}. \quad (60)$$

The same projection is applied at initialization:

$$\widehat{\mathbf{J}}_0 := \text{proj}_{\mathcal{J}} \left[\mathbf{J} \left(\widehat{\boldsymbol{\theta}}_{J,0}^{\text{rls}} \right) \right], \quad (61)$$

where $\text{proj}_{\mathcal{J}}$ denotes the eigenvalue-clipping operation in Eqs. (58)–(60). Thus, a valid inertia estimate is available for the initial control computation. The corresponding projected inertia-parameter vector is

$$\widehat{\boldsymbol{\theta}}_{J,k} := \begin{bmatrix} \widehat{J}_{11,k} & \widehat{J}_{22,k} & \widehat{J}_{33,k} & \widehat{J}_{12,k} & \widehat{J}_{13,k} & \widehat{J}_{23,k} \end{bmatrix}^\top. \quad (62)$$

The unconstrained estimate $\widehat{\boldsymbol{\theta}}_{J,k}^{\text{rls}}$ is retained as the RLS state, whereas the projected inertia matrix $\widehat{\mathbf{J}}_k$ is used for prediction and control.

The projection guarantees

$$\left\| \widehat{\mathbf{J}}_k^{-1} \right\|_2 \leq \frac{1}{\underline{J}}, \quad \kappa_2 \left(\widehat{\mathbf{J}}_k \right) \leq \frac{\bar{J}}{\underline{J}}. \quad (63)$$

Algorithm 1 Online Full-Inertia Identification

Require: ω_{k-1} , ω_k , τ_{k-1} , $\hat{\theta}_{J,k-1}^{\text{rls}}$, and $\mathbf{P}_{k-1}$

- 1: Construct $\mathbf{b}_{J,k}$ using Eq. (42)
 - 2: Construct $\Phi_{J,k}$ using Eq. (43)
 - 3: Compute $\varepsilon_{J,k}$ using Eq. (46)
 - 4: Compute β_k using Eqs. (48) and (49)
 - 5: Update $\mathbf{P}_k$ and $\hat{\theta}_{J,k}^{\text{rls}}$ using Eqs. (53) and (54)
 - 6: Reconstruct $\hat{\mathbf{J}}_k^{\text{rls}}$
 - 7: Project $\hat{\mathbf{J}}_k^{\text{rls}}$ onto $\mathcal{J}$ to obtain $\hat{\mathbf{J}}_k$
 - 8: Compute and store a Cholesky factorization of $\hat{\mathbf{J}}_k$
 - 9: Use $\hat{\mathbf{J}}_k$ in the quaternion sampled predictor
-

Hence, an indefinite or zero unconstrained estimate still produces a bounded, nonsingular matrix. In particular, a zero estimate is projected onto $\underline{\mathbf{J}}\mathbf{I}_3$.

The forward predictor does not explicitly compute $\hat{\mathbf{J}}_k^{-1}$. Instead, the required angular acceleration is obtained by solving

$$\hat{\mathbf{J}}_k \hat{\dot{\omega}} = -\hat{\omega}^\times \hat{\mathbf{J}}_k \hat{\omega} + \tau. \quad (64)$$

A Cholesky factorization of $\hat{\mathbf{J}}_k$ is computed once per control update and reused during predictor propagation.

The transition ending at t_k is processed before the control computation at step k . Therefore, $\hat{\mathbf{J}}_k$ is available when the next torque command is computed.

No external probing signal is added for identification. The regressors are generated by the reference-tracking and corrective torques produced by the closed-loop PCAC system. Variable-rate forgetting facilitates readaptation after an inertia change, but it does not generate excitation. Complete parameter convergence is therefore not asserted without persistent excitation.

V. Quaternion Sampled Predictor

The identified inertia matrix provides an inverse-dynamics model, whereas predictive control requires a forward state transition. This section constructs a sampled predictor using the projected estimate $\hat{\mathbf{J}}_k$. At control instant k , $\hat{\mathbf{J}}_k$ is held fixed over the prediction horizon, and the torque is assumed constant over each sampling interval.

A. Structure-Preserving Integration

For $\omega, \tau \in \mathbb{R}^3$, define $\hat{\mathbf{f}}_{\omega,k}(\omega, \tau)$ as the solution of

$$\hat{\mathbf{J}}_k \hat{\mathbf{f}}_{\omega,k}(\omega, \tau) = \tau - \omega^\times \hat{\mathbf{J}}_k \omega. \quad (65)$$

Since $\widehat{\mathbf{J}}_k \in \mathcal{J}$, this solution exists and is unique. It is evaluated using the Cholesky factorization of $\widehat{\mathbf{J}}_k$; the matrix inverse is not formed explicitly.

Let $N_s \in \mathbb{N}$ denote the number of internal integration steps and define

$$h := \frac{T_s}{N_s}. \quad (66)$$

For a generic initial state $(\mathbf{q}_0, \omega_0) = (\mathbf{q}, \omega)$, the angular velocity is propagated using the explicit midpoint method:

$$\omega_{\ell+\frac{1}{2}} = \omega_\ell + \frac{h}{2} \widehat{\mathbf{f}}_{\omega,k}(\omega_\ell, \tau), \quad (67)$$

240

$$\omega_{\ell+1} = \omega_\ell + h \widehat{\mathbf{f}}_{\omega,k}(\omega_{\ell+\frac{1}{2}}, \tau), \quad (68)$$

for $\ell = 0, \dots, N_s - 1$.

For $\xi \in \mathbb{R}^3$, define

$$\text{Exp}_q(\xi) := \begin{bmatrix} \cos \rho \\ \text{sinc}(\rho)\xi \end{bmatrix}, \quad \rho := \|\xi\|_2, \quad (69)$$

where $\text{sinc}(\rho) = \sin(\rho)/\rho$ for $\rho \neq 0$ and $\text{sinc}(0) = 1$. The quaternion update is

$$\mathbf{q}_{\ell+1} = \mathbf{q}_\ell \otimes \text{Exp}_q\left(\frac{h}{2}\omega_{\ell+\frac{1}{2}}\right). \quad (70)$$

Since $\text{Exp}_q(\xi)$ is a unit quaternion,

$$\|\mathbf{q}_{\ell+1}\|_2 = \|\mathbf{q}_\ell\|_2. \quad (71)$$

Thus, the update preserves the unit-quaternion constraint in exact arithmetic. A single normalization after each complete sampling interval may be used to compensate for floating-point roundoff.

The angular-velocity update is second-order accurate. The multiplicative quaternion update is exact for constant angular velocity over a substep and, when evaluated using the midpoint angular velocity, gives a second-order structure-preserving approximation of the attitude motion.

B. Adaptive Nonlinear Sampled Map

Recall the attitude state $\mathbf{x}$ defined in (16). Starting from $(\mathbf{q}_0, \omega_0) = (\mathbf{q}, \omega)$, apply Eqs. (67)–(70) for $\ell = 0, \dots, N_s - 1$.

The resulting sampled predictor is

$$\widehat{\mathbf{F}}_{s,k}(\mathbf{x}, \tau) := \begin{bmatrix} \mathbf{q}_{N_s} \\ \omega_{N_s} \end{bmatrix}, \quad (72)$$

so that

$$\mathbf{x}^+ = \widehat{\mathbf{F}}_{s,k}(\mathbf{x}, \tau). \quad (73)$$

The subscript s denotes the sampled state transition. The map $\widehat{\mathbf{F}}_{s,k}$ is a fixed-step approximation of the zero-order-hold rigid-body flow; it is not claimed to be an exact discretization of the complete nonlinear dynamics. It preserves the quaternion constraint and uses the current projected inertia estimate throughout the prediction horizon.

Setting $N_s = 1$ gives one midpoint step per controller sampling interval. Additional internal steps can be used when the sampling period, angular velocity, or applied torque requires finer numerical propagation. The sampled map is reconstructed whenever a new inertia estimate becomes available.

The next section uses $\widehat{\mathbf{F}}_{s,k}$ to construct the local Jacobian-frozen affine predictor employed by PCAC.

VI. Predictive Cost Adaptive Control

The adaptive sampled map $\widehat{\mathbf{F}}_{s,k}$ is nonlinear. Direct propagation over a finite horizon would lead to a nonlinear optimization problem. Predictive cost adaptive control (PCAC) uses a local affine approximation of this map. The approximation is updated at each sampling instant and held fixed over the prediction horizon.

A. Local Quaternion Coordinates

The quaternion has four components but only three independent degrees of freedom. The sampled predictor is therefore expressed in local six-dimensional coordinates.

Let

$$\mathbf{R}_k := \mathbf{R}(\mathbf{q}_k). \quad (74)$$

For any $\mathbf{x} = [\mathbf{q}^\top \ \boldsymbol{\omega}^\top]^\top$ near $\mathbf{x}_k$, define

$$\boldsymbol{\zeta} := C_k(\mathbf{x}) := \begin{bmatrix} \boldsymbol{\vartheta} \\ \boldsymbol{\omega}_{\text{loc}} \end{bmatrix} \in \mathbb{R}^6, \quad (75)$$

where

$$\boldsymbol{\vartheta} := \mathcal{L}(\mathbf{q}_k^* \otimes \mathbf{q}), \quad (76)$$

and

$$\boldsymbol{\omega}_{\text{loc}} := \mathbf{R}_k^\top \mathbf{R}(\mathbf{q}) \boldsymbol{\omega}. \quad (77)$$

The map $\mathcal{L}$ denotes the principal rotation-vector construction used in Eq. (21). The quaternion argument is selected with nonnegative scalar part before evaluating $\mathcal{L}$.

The vector $\boldsymbol{\omega}_{\text{loc}}$ expresses the angular velocity in the fixed body frame associated with $\mathbf{q}_k$. Predicted and desired

angular velocities are therefore represented in the same frame.

At the current state,

$$\boldsymbol{\zeta}_k := C_k(\mathbf{x}_k) = \begin{bmatrix} \mathbf{0}_3 \\ \boldsymbol{\omega}_k \end{bmatrix}. \quad (78)$$

For $\boldsymbol{\zeta} = [\boldsymbol{\vartheta}^\top \boldsymbol{\omega}_{\text{loc}}^\top]^\top$ with $\|\boldsymbol{\vartheta}\|_2 < \pi$, define

$$\mathbf{q}(\boldsymbol{\vartheta}) := \mathbf{q}_k \otimes \text{Exp}_q\left(\frac{1}{2}\boldsymbol{\vartheta}\right). \quad (79)$$

The inverse coordinate map is

$$C_k^{-1}(\boldsymbol{\zeta}) := \begin{bmatrix} \mathbf{q}(\boldsymbol{\vartheta}) \\ \mathbf{R}^\top(\mathbf{q}(\boldsymbol{\vartheta})) \mathbf{R}_k \boldsymbol{\omega}_{\text{loc}} \end{bmatrix}. \quad (80)$$

Using Eqs. (72) and (80), define the nonlinear sampled map in local coordinates:

$$\widehat{\mathbf{F}}_{\boldsymbol{\zeta},k}(\boldsymbol{\zeta}, \boldsymbol{\tau}) := C_k \left[\widehat{\mathbf{F}}_{s,k} \left(C_k^{-1}(\boldsymbol{\zeta}), \boldsymbol{\tau} \right) \right]. \quad (81)$$

The local representation is valid while the predicted attitude remains within the principal rotation domain relative to $\mathbf{q}_k$.

B. Jacobian-Frozen Affine Predictor

At step k , the current torque $\boldsymbol{\tau}_k$ is known. Evaluate the Jacobians of $\widehat{\mathbf{F}}_{\boldsymbol{\zeta},k}$ at the current operating point:

$$\mathbf{A}_k := \left. \frac{\partial \widehat{\mathbf{F}}_{\boldsymbol{\zeta},k}}{\partial \boldsymbol{\zeta}} \right|_{(\boldsymbol{\zeta}_k, \boldsymbol{\tau}_k)} \in \mathbb{R}^{6 \times 6}, \quad (82)$$

$$\mathbf{B}_k := \left. \frac{\partial \widehat{\mathbf{F}}_{\boldsymbol{\zeta},k}}{\partial \boldsymbol{\tau}} \right|_{(\boldsymbol{\zeta}_k, \boldsymbol{\tau}_k)} \in \mathbb{R}^{6 \times 3}. \quad (83)$$

Define the affine term

$$\mathbf{c}_k := \widehat{\mathbf{F}}_{\boldsymbol{\zeta},k}(\boldsymbol{\zeta}_k, \boldsymbol{\tau}_k) - \mathbf{A}_k \boldsymbol{\zeta}_k - \mathbf{B}_k \boldsymbol{\tau}_k. \quad (84)$$

The Jacobian-frozen predictor is

$$\boldsymbol{\zeta}_{i+1|k} = \mathbf{c}_k + \mathbf{A}_k \boldsymbol{\zeta}_{i|k} + \mathbf{B}_k \boldsymbol{\tau}_{i|k}, \quad i = 0, \dots, N-1. \quad (85)$$

The matrices $\mathbf{A}_k$, $\mathbf{B}_k$, and $\mathbf{c}_k$ are held fixed over the horizon. They are recomputed after the next state measurement and inertia update.

288 The Jacobians may be evaluated analytically or by automatic differentiation. A centered-difference implementation
 289 is

$$[\mathbf{A}_k]_{:j} \approx \frac{\widehat{\mathbf{F}}_{\zeta,k}(\zeta_k + \delta_{\zeta,j} \mathbf{e}_j, \tau_k) - \widehat{\mathbf{F}}_{\zeta,k}(\zeta_k - \delta_{\zeta,j} \mathbf{e}_j, \tau_k)}{2\delta_{\zeta,j}}, \quad (86)$$

290 for $j = 1, \dots, 6$, and

$$[\mathbf{B}_k]_{:j} \approx \frac{\widehat{\mathbf{F}}_{\zeta,k}(\zeta_k, \tau_k + \delta_{\tau,j} \mathbf{e}_j) - \widehat{\mathbf{F}}_{\zeta,k}(\zeta_k, \tau_k - \delta_{\tau,j} \mathbf{e}_j)}{2\delta_{\tau,j}}, \quad (87)$$

291 for $j = 1, \dots, 3$, where $\mathbf{e}_j$ is the corresponding Euclidean unit vector. The perturbation magnitudes $\delta_{\zeta,j} > 0$ and
 292 $\delta_{\tau,j} > 0$ are selected according to the scale of each variable.

293 C. Predictive Optimization

294 Let $N \geq 2$ denote the prediction horizon. Following the PCAC timing convention, initialize

$$\zeta_{0|k} = \zeta_k, \quad \tau_{0|k} = \tau_k. \quad (88)$$

295 The current torque is known. The decision variables are

$$\tau_{1|k}, \dots, \tau_{N-1|k}. \quad (89)$$

296 Let

$$\mathbf{x}_{d,i|k} := \begin{bmatrix} \mathbf{q}_{d,i|k} \\ \omega_{d,i|k} \end{bmatrix} \quad (90)$$

297 denote the desired state at prediction step i . Its local representation is

$$\zeta_{d,i|k} := C_k(\mathbf{x}_{d,i|k}). \quad (91)$$

298 Define the predicted torque increments

$$\Delta \tau_{i|k} := \tau_{i|k} - \tau_{i-1|k}, \quad i = 1, \dots, N-1. \quad (92)$$

299 For $\mathbf{Q}_i \succeq \mathbf{0}$ and $\mathbf{R}_\tau \succ \mathbf{0}$, solve

$$\begin{aligned}
& \underset{\substack{\zeta_{1|k}, \dots, \zeta_{N|k}, \\ \tau_{1|k}, \dots, \tau_{N-1|k}}}{\text{minimize}} & \frac{1}{2} \sum_{i=1}^N \|\zeta_{i|k} - \zeta_{d,i|k}\|_{\mathbf{Q}_i}^2 + \frac{1}{2} \sum_{i=1}^{N-1} \|\Delta \tau_{i|k}\|_{\mathbf{R}_\tau}^2 \\
& \text{subject to} & \zeta_{i+1|k} = \mathbf{c}_k + \mathbf{A}_k \zeta_{i|k} + \mathbf{B}_k \tau_{i|k}, \\
& & i = 0, \dots, N-1, \\
& & \tau_{\min} \leq \tau_{i|k} \leq \tau_{\max}, \quad i = 1, \dots, N-1, \\
& & \Delta \tau_{\min} \leq \Delta \tau_{i|k} \leq \Delta \tau_{\max}, \quad i = 1, \dots, N-1.
\end{aligned} \tag{93}$$

300 Here, $\|\mathbf{v}\|_{\mathbf{M}}^2 := \mathbf{v}^\top \mathbf{M} \mathbf{v}$. The terminal tracking weight is included in $\mathbf{Q}_N$.

301 Since the predictor is affine and the constraints are linear, Eq. (93) is a convex quadratic program (QP). The
302 positive-definite torque-increment weight gives a unique optimal input sequence.

303 Let $\tau_{1|k}^\star$ denote the first optimized torque. The PCAC timing convention applies

$$304 \quad \tau_{k+1} = \tau_{1|k}^\star. \tag{94}$$

304 The initial torque τ_0 is prescribed.

305 D. Online Implementation

306 At each sampling instant, the transition ending at t_k is first used to update and project the inertia estimate. The
307 nonlinear sampled map is then constructed using $\widehat{\mathbf{J}}_k$. Its local Jacobians are evaluated at (ζ_k, τ_k) , and the resulting
308 affine model is frozen over the prediction horizon. Finally, the quadratic program in Eq. (93) is solved to compute τ_{k+1} .

309 The nonlinear sampled predictor and its Jacobians are updated once per sampling instant. The finite-horizon control
310 computation remains a quadratic program; no nonlinear optimization is performed online.

311 VII. Numerical Evaluation

312 A. General Setup

313 The attitude is represented by a scalar-first Hamilton quaternion. The controller combines a geometry-informed
314 nonlinear predictor with PCAC, and the implementation avoids explicit matrix inversion. The proposed, no-forgetting,
315 diagonal-adaptive, and fixed-inertia modes use their current projected or retained inertia model, whereas the exact-inertia
316 benchmark uses the true instantaneous inertia. As detailed in Section VI.D, the completed transition is processed before
317 control, the updated projected estimate is used for prediction, and the applied convention is $\tau_{k+1} = \tau_{1|k}^\star$.

318 The common cost weights are

$$\mathbf{Q}_i = \mathbf{Q} = \text{diag}(40, 40, 40, 5, 5, 5), \quad i = 1, \dots, N-1, \quad \mathbf{Q}_N = 10\mathbf{Q}, \quad \mathbf{R}_\tau = 0.2\mathbf{I}_3. \quad (95)$$

319 The componentwise input constraints are

$$-0.15\mathbf{1}_3 \leq \boldsymbol{\tau}_k \leq 0.15\mathbf{1}_3 \quad \text{Nm}, \quad (96)$$

$$-0.03\mathbf{1}_3 \leq \Delta\boldsymbol{\tau}_k \leq 0.03\mathbf{1}_3 \quad \text{Nm} \quad (97)$$

320 per sampling interval, where $\mathbf{1}_3$ is the three-dimensional vector of ones. For numerical robustness, each case augments
321 the QP with a nonnegative slack vector $\mathbf{s}_{\omega,i|k}$ satisfying

$$\begin{aligned} -\omega_{\max}\mathbf{1}_3 - \mathbf{s}_{\omega,i|k} &\leq \boldsymbol{\omega}_{\text{loc},i|k} \leq \omega_{\max}\mathbf{1}_3 + \mathbf{s}_{\omega,i|k}, \\ \mathbf{s}_{\omega,i|k} &\geq \mathbf{0}_3, \quad i = 1, \dots, N. \end{aligned} \quad (98)$$

322 The cost in Eq. (93) is augmented by

$$\frac{r_s}{2} \sum_{i=1}^N \|\mathbf{s}_{\omega,i|k}\|_2^2, \quad \omega_{\max} = 1.5 \text{ rad s}^{-1}, \quad r_s = 10^6. \quad (99)$$

323 This soft bound is a numerical augmentation; the torque and torque-increment constraints remain hard. The sampled
324 nonlinear predictor uses $N_s = 2$ internal midpoint steps per sampling interval. The true plant is propagated by
325 fourth-order Runge–Kutta integration with 20 substeps per sampling interval.

326 The two constant endpoint inertia matrices used below, both expressed in kg m^2 , are

$$\begin{aligned} \mathbf{J}_A &= \begin{bmatrix} 0.03445 & -0.00263 & -0.00414 \\ -0.00263 & 0.04048 & -0.00448 \\ -0.00414 & -0.00448 & 0.05307 \end{bmatrix}, \\ \mathbf{J}_B &= \begin{bmatrix} 0.03107 & 0.00427 & 0.01169 \\ 0.00427 & 0.04703 & 0.00152 \\ 0.01169 & 0.00152 & 0.05690 \end{bmatrix}. \end{aligned} \quad (100)$$

Cases 2 and 3 use the common multiaxis reference

$$\boldsymbol{\omega}_d(t) = \begin{bmatrix} 0.45 \sin(0.35t) \\ 0.35 \sin(0.50t + \pi/3) \\ 0.30 \sin(0.65t + \pi/6) \end{bmatrix} \text{ rad s}^{-1}. \quad (101)$$

The initial desired quaternion is $\mathbf{q}_d(0) = [1, 0, 0, 0]^\top$, and $\mathbf{q}_d(t)$ is generated by integrating the quaternion kinematics consistently with $\boldsymbol{\omega}_d(t)$. Within each case, all compared controllers use identical initial conditions, reference trajectories, constraints, and measurement-noise sequences.

B. Case 1: Monte Carlo Large-Angle Stabilization

1. Simulation Setup

Case 1 evaluates large-angle stabilization under uncertain initial conditions and uncertain initial inertia estimates. The study contains 200 paired Monte Carlo trials with deterministic seeds from 260701 to 260900. Each trial lasts 12 s with sampling period $T_s = 0.02$ s and prediction horizon $N = 20$. The reference is the fixed attitude $\mathbf{q}_d = [1, 0, 0, 0]^\top$ with angular velocity $\boldsymbol{\omega}_d = \mathbf{0}$. The true full inertia is the constant matrix $\mathbf{J}_A$. Measurement noise is zero, and no external probing signal is applied.

For each trial, the initial attitude-error angle is sampled uniformly from 60° to 170° about an isotropically distributed axis. Each initial angular-velocity component is sampled from $[-0.25, 0.25]$ rad s⁻¹. The initial inertia estimate is a random symmetric positive-definite matrix with eigenvalues sampled from $[0.015, 0.080]$ kg m² and random orthogonal principal axes. Within each paired trial, all controllers use the same initial state, initial inertia realization, reference, constraints, and noise sequence.

The Case 1 estimator parameters are

$$\eta = 0.01, \quad \sigma_\varepsilon = 2 \times 10^{-4}, \quad N_f = 25, \quad \gamma = 5, \quad \mathbf{P}_0 = 10^4 \mathbf{I}_{n_\theta}. \quad (102)$$

Here, $n_\theta = 6$ for full-inertia estimation and $n_\theta = 3$ for diagonal-only estimation. These values were not selected through a Case 1-specific sensitivity study and are not claimed to be optimal.

Four controllers are compared. The proposed controller combines full-inertia VRF-RLS, positive-definite inertia projection, and geometry-informed PCAC. It estimates all six independent entries of the symmetric inertia matrix. The diagonal-adaptive controller uses the same VRF-RLS update, inertia projection, and PCAC formulation, but estimates only the three diagonal inertia entries and constrains the products of inertia to zero. The fixed-inertia controller retains the randomly sampled initial projected estimate throughout the trial. The exact-inertia controller uses the true $\mathbf{J}_A$ in the

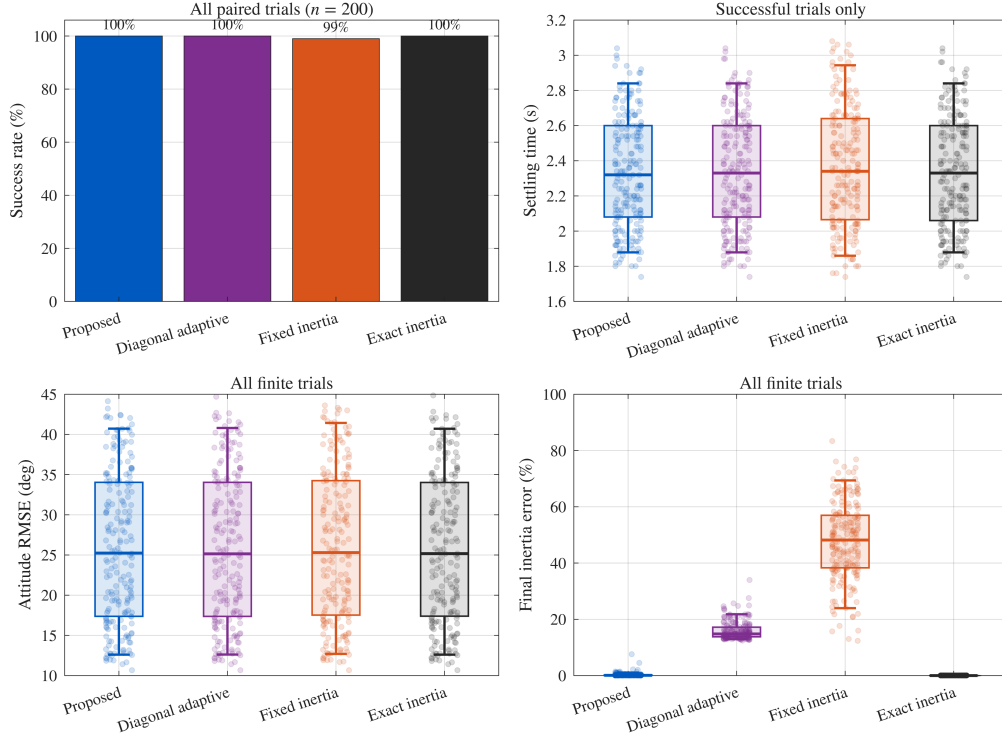


Fig. 1 Case 1 distributions of success, settling time, tracking error, and final inertia error across 200 paired Monte Carlo trials for four controllers.

same predictor and PCAC formulation. The fixed- and exact-inertia controllers do not perform an RLS update. The exact-inertia controller is an ideal simulation benchmark and is not implementable when the true inertia is unknown.

A trial is successful when $\|\mathbf{e}_R\|$ remains below 2° and $\|\mathbf{e}_\omega\|$ remains below 0.05 rad s^{-1} throughout the final 2 s. The trajectory must also remain finite, satisfy the constraints and QP residual checks, and contain no failed QP update. Success rates use all 200 trials as the denominator. Settling-time statistics include successful trials only, whereas the other distribution summaries include every finite trial.

2. Results and Discussion

Figure 1 presents the aggregate Monte Carlo results, and Table 1 gives the corresponding numerical summaries. Root-mean-square error (RMSE) is evaluated over the full simulation interval, and the interquartile range (IQR) describes distribution spread. Success rates use all trials; the settling-time sample sizes for the proposed, diagonal-adaptive, fixed-inertia, and exact-inertia controllers are 200, 200, 198, and 200, respectively. In the figure, points denote trials, boxes span the IQR, center lines show medians, and whiskers mark the 5th–95th percentiles.

The proposed, diagonal-adaptive, and exact-inertia controllers succeed in all 200 trials. The fixed-inertia controller succeeds in 198 trials. Its two unsuccessful trials have different causes. Trial 93 does not satisfy the settling requirement and contains no QP failure. Trial 132 settles but contains two failed QP updates. All controller modes complete the

Table 1 Case 1 Monte Carlo performance over 200 paired trials.

Metric	Proposed	Diagonal adaptive	Fixed inertia	Exact inertia
	200/200	200/200	198/200	200/200
Success, trials (%)	(100.0)	(100.0)	(99.0)	(100.0)
Settling time, median [IQR] (s)	2.32 [0.52]	2.33 [0.52]	2.34 [0.58]	2.33 [0.54]
Attitude RMSE, median [IQR] (°)	25.241 [16.669]	25.145 [16.676]	25.300 [16.724]	25.173 [16.634]
Angular-rate RMSE, median [IQR] (rad s ⁻¹)	0.4951 [0.1768]	0.4956 [0.1810]	0.4974 [0.1840]	0.4937 [0.1786]
Final inertia error, median [IQR] (%)	0.09 [0.12]	14.80 [3.41]	48.18 [18.68]	0.00 [0.00]
QP failures	0	0	2	0

Brackets contain interquartile ranges. Settling-time statistics are conditional on success. The QP-failure row counts failed QP updates rather than unsuccessful trials.

study with zero RLS update failures and zero projection fallbacks.

The settling-time and tracking-error distributions are similar across the four controllers. In particular, the proposed and exact-inertia controllers have nearly identical settling-time distributions. The median settling times are approximately 2.3 s, which is short relative to the 12 s simulation duration. By contrast, the median attitude RMSE values are approximately 25°. These RMSE values include the complete response and are dominated by the sampled initial attitude errors of up to 170°; they do not represent the final stabilized error.

The similar tracking distributions show that the closed-loop response in this stabilization study is relatively insensitive to the remaining model error for most sampled trials. This similarity does not imply that the controllers identify the inertia equally well or that comparable inertia errors would be harmless for other references, constraints, or inertia profiles.

The proposed estimator has a median final relative inertia error of 0.09%. Its 95th-percentile error is 1.07%, while the maximum is 7.63%. The estimator therefore achieves highly accurate final identification in most trials, although the accuracy is not uniform across the complete sample. The identification results are consistent with the transient stabilization motion providing sufficient excitation for accurate full-inertia estimation in most realizations despite the absence of an external probing signal.

The diagonal-adaptive estimator retains a median final inertia error of 14.80%. This persistent error is consistent with its inability to represent the products of inertia. The fixed-inertia controller retains a median error of 48.18% and reaches a maximum of 83.35% because it cannot correct its randomly sampled initial model mismatch. The small difference from the exact-inertia benchmark suggests that model uncertainty is not the dominant limitation for most trials in this stabilization study.

Figure 2 shows trial 24, with seed 260724. This realization is selected deterministically as the successful proposed trial nearest the median proposed settling time. The selection rule concerns settling time and does not select for favorable identification performance.

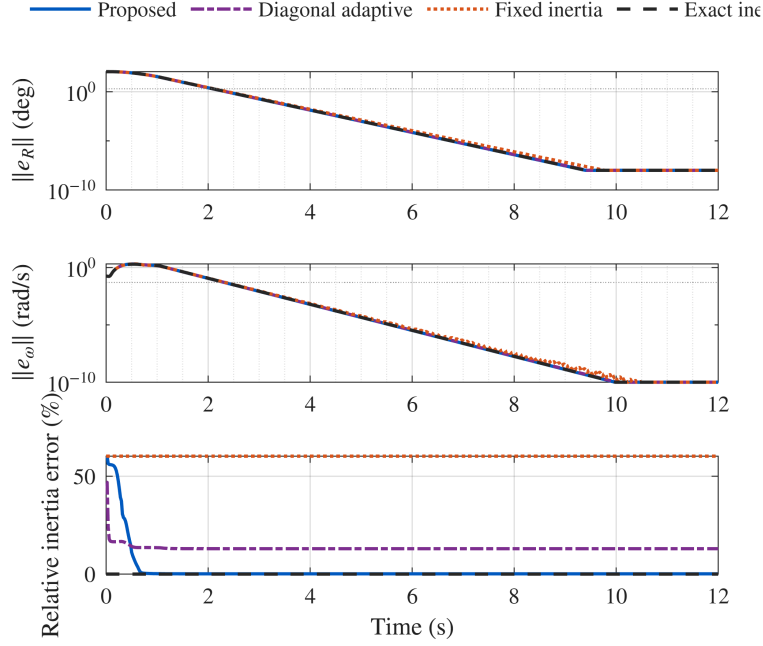


Fig. 2 Representative Case 1 tracking and inertia-estimation histories for trial 24, selected by median settling time among successful proposed-controller trials in the study.

The representative tracking curves nearly overlap. The settling times range from 2.32 s to 2.34 s, and the full-interval tracking RMSE values are also similar. The inertia-error histories differ substantially. The proposed final error is 0.137%, compared with 12.979% for diagonal adaptation and 60.248% for fixed inertia. This realization illustrates why stabilization and identification accuracy must be assessed separately.

Figure 3 provides a geometric view of the same proposed-controller realization. The circle, square, and star mark the initial, final, and desired body-axis directions, respectively; dotted and solid triads denote the initial and final body frames.

The sphere plot illustrates the large-angle geometric maneuver but represents only one trial. A single body-axis trajectory does not uniquely describe the complete attitude because rotation about that axis is not shown. The aggregate results in Fig. 1 and Table 1 provide the quantitative Monte Carlo evidence.

Overall, Case 1 demonstrates successful large-angle stabilization in every proposed-controller trial, tracking close to the exact-inertia benchmark, and accurate full-inertia identification in most trials. Restricting adaptation to diagonal inertia produces persistent identification error, while retaining the initial inertia estimate causes large model error and a small loss of stabilization robustness.

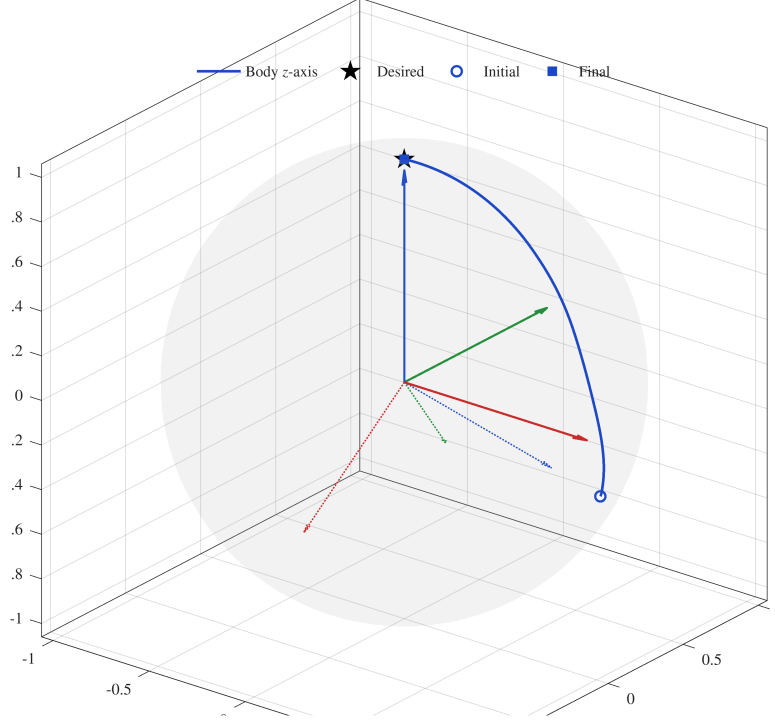


Fig. 3 Body z -axis trajectory and initial/final body-frame triads for the representative Case 1 proposed-controller realization, showing convergence to the desired attitude.

C. Case 2: Rapid Full-Inertia Change

1. Simulation Setup

The simulation duration is 30 s, the sampling period is $T_s = 0.02$ s, and the prediction horizon is $N = 20$. The common multi-axis reference defined in Section VII.A is used, and measurement noise is set to zero. The initial attitude error is a 25° rotation about the unit axis proportional to $[1, -0.5, 0.8]^\top$, and the initial angular-velocity error is $\mathbf{e}_{\omega,0} = [0.05, -0.04, 0.03]^\top \text{ rad s}^{-1}$.

The true inertia changes smoothly from $\mathbf{J}_A$ to $\mathbf{J}_B$ during 15–15.25 s. During this interval, the plant dynamics include the effect of the inertia rate:

$$\mathbf{J}(t)\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{J}(t)\boldsymbol{\omega}) - \dot{\mathbf{J}}(t)\boldsymbol{\omega}. \quad (103)$$

At each control update, the predictor is constructed from the current inertia model, which is held fixed over the complete prediction horizon. The inertia-rate term is absent from the interval-constant identification model and therefore contributes to the regression disturbance $\mathbf{v}_{J,k}$ during the transition.

Four controllers are compared. The proposed controller combines full-inertia VRF-RLS, positive-definite inertia projection, and geometry-informed PCAC. The no-forgetting controller uses the same estimator and control formulation with $\beta_k = 1$. The fixed-inertia controller retains the initial projected inertia estimate throughout the simulation. The exact-inertia controller uses the true instantaneous inertia $\mathbf{J}(t_k)$ in the same predictor and PCAC formulation. It is an

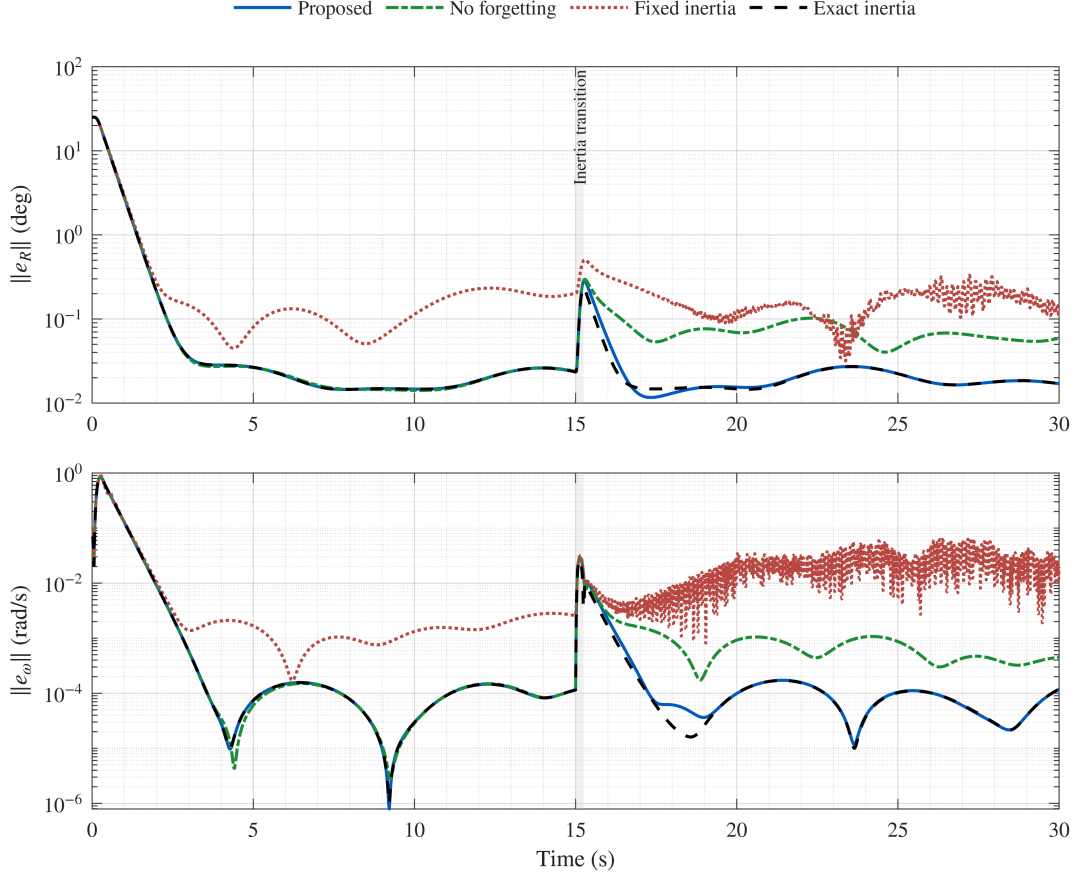


Fig. 4 Case 2 tracking errors during the rapid inertia transition, comparing adaptive, fixed-inertia, and exact-inertia controllers before and after the change.

ideal causal simulation benchmark and is not implementable when the true inertia is unknown.

The VRF sensitivity study selected exactly one parameter combination:

$$\eta = 0.1, \quad \sigma_{\varepsilon} = 10^{-4}, \quad N_f = 25, \quad \gamma = 5. \quad (104)$$

The selected run had zero QP failures, zero RLS update failures, and zero projection fallbacks.

2. Results and Discussion

Table 2 reports the peak attitude-error norm after the inertia transition and the RMSE values over the final 5 s. Here, $\mathbf{e}_R$ denotes the attitude error and $\mathbf{e}_\omega$ denotes the angular-velocity error. Recovery time is omitted because the current threshold-based metric gives 0.01 s for every controller and therefore does not distinguish their responses.

Figure 4 shows that the proposed controller approaches the tracking performance of the exact-inertia benchmark after the transition. Their final attitude and angular-velocity RMSE values are nearly identical. The remaining difference is concentrated during and immediately after the inertia change, when the proposed controller must estimate the new

Table 2 Case 2 tracking and identification performance after the rapid full-inertia transition.

Controller	Peak $\ \mathbf{e}_R\ $ ($^\circ$)	Final $\mathbf{e}_R$ RMSE ($^\circ$)	Final $\mathbf{e}_\omega$ RMSE (rad s^{-1})	Final relative inertia error (%)
Proposed	0.287	0.018	7.482×10^{-5}	0.04
No forgetting	0.296	0.060	4.484×10^{-4}	28.64
Fixed inertia	0.505	0.199	3.253×10^{-2}	51.41
Exact inertia	0.237	0.018	7.470×10^{-5}	0

inertia while the exact-inertia controller already has access to it.

Variable-rate forgetting gives only a small reduction in the peak attitude error relative to no-forgetting RLS, from 0.296° to 0.287° . Its effect is clearer in the final tracking and identification results. The proposed controller reaches a final inertia error of 0.04%, compared with 28.64% for no-forgetting RLS. Its final angular-velocity RMSE is also approximately six times smaller. These results indicate near-complete reidentification in the proposed case.

Figure 5 shows that the proposed estimates approach the true diagonal and off-diagonal inertia entries after the shaded transition interval. The minimum instantaneous forgetting factor is

$$\min_k \beta_k^{-1} \approx 0.725. \quad (105)$$

Reducing β_k^{-1} lowers the influence of pre-transition data when the inertia changes. This allows the estimator to identify the new inertia more completely than ordinary RLS. With no forgetting, old data retain their full weight, and the post-transition estimate remains biased.

The fixed-inertia controller cannot correct its model mismatch. Its retained estimate has an error of approximately 42.72% relative to $\mathbf{J}_A$ and 51.41% relative to $\mathbf{J}_B$. Consequently, it has the largest peak and final tracking errors and retains oscillatory residual errors. The repeated fixed-inertia simulation is identical to floating-point precision. The oscillations are therefore reproducible rather than stochastic, and their persistence is consistent with the retained model mismatch. Because this controller starts with a poor inertia estimate, it serves as a fixed poor-initial-estimate baseline; it does not isolate only the effect of disabling adaptation after the inertia transition.

Overall, online full-inertia adaptation allows the prediction model to follow the changed plant dynamics. Variable-rate forgetting improves the post-transition inertia estimate by reducing the influence of obsolete data, and the resulting prediction model brings the proposed controller close to the exact-inertia benchmark.

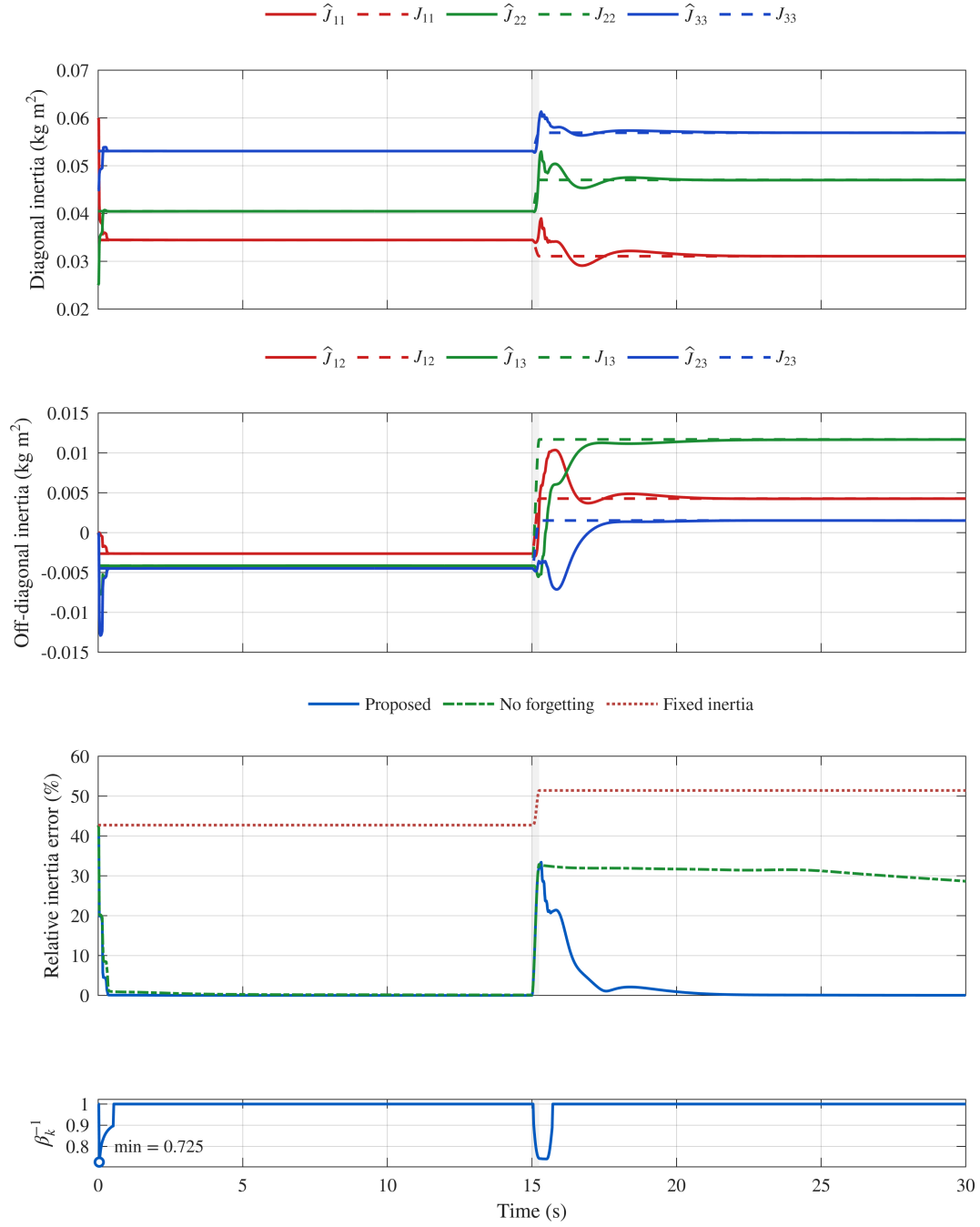


Fig. 5 Case 2 inertia estimates, relative identification error, and instantaneous forgetting factor during and after the rapid transition in the true inertia.

D. Case 3: Continuously Varying Full Inertia

1. Simulation Setup

Case 3 examines tracking and identification when the true full inertia varies slowly and continuously. The simulation duration is 60 s, the sampling period is $T_s = 0.02$ s, and the prediction horizon is $N = 20$. The common multiaxis

reference defined in Section VII.A is used, measurement noise is zero, and no external probing signal is applied. The initial attitude error is a 20° rotation about the unit axis proportional to $[0.7, 1, -0.4]^\top$, and the initial angular-velocity error is $\mathbf{e}_{\omega,0} = [-0.03, 0.04, -0.02]^\top \text{ rad s}^{-1}$.

The true inertia is

$$\mathbf{J}(t) = \mathbf{U}_A \text{diag}(j_1(t), j_2(t), j_3(t)) \mathbf{U}_A^\top, \quad (106)$$

where $\mathbf{U}_A$ is the fixed orthogonal matrix of principal axes associated with $\mathbf{J}_A$. The principal inertias, in kg m^2 , are

$$j_1(t) = 0.032 [1 + 0.12 \sin(0.10t)], \quad (107)$$

$$j_2(t) = 0.041 \left[1 + 0.10 \sin \left(0.08t + \frac{2\pi}{3} \right) \right], \quad (108)$$

$$j_3(t) = 0.055 \left[1 + 0.08 \sin \left(0.06t + \frac{4\pi}{3} \right) \right]. \quad (109)$$

The true plant includes the $\dot{\mathbf{J}}(t)\boldsymbol{\omega}$ term. The identification model remains interval-constant, so this inertia-rate contribution enters $\nu_{J,k}$ as model mismatch. All controllers use identical initial conditions, references, constraints, and measurement-noise sequences.

Three controllers are compared. The proposed controller combines full-inertia VRF-RLS, positive-definite inertia projection, and geometry-informed PCAC. The no-forgetting controller uses the same estimator and control formulation with $\beta_k = 1$. The exact-inertia controller uses the true instantaneous inertia $\mathbf{J}(t_k)$ in the same predictor and PCAC formulation. It provides an ideal causal simulation benchmark but is not implementable when the true inertia is unknown.

The proposed estimator uses

$$\eta = 0.01, \quad \sigma_\varepsilon = 2 \times 10^{-4}, \quad N_f = 25, \quad \gamma = 5, \quad \mathbf{P}_0 = 10^4 \mathbf{I}_6. \quad (110)$$

$N_f = 25$ is the residual-normalization window used to determine VRF activation and corresponds to 0.5 s at $T_s = 0.02$ s. It is not the RLS parameter-memory length; the influence of previous parameter data is governed by the covariance recursion and the forgetting factor. No Case 3-specific sensitivity study was performed, so these parameters are not claimed to be optimal.

2. Results and Discussion

Figure 6 compares the tracking errors, and Table 3 summarizes the full-interval results. The proposed and no-forgetting trajectories are visually almost indistinguishable. Their attitude and angular-velocity RMSE values agree to the reported precision, and their mean inertia errors differ by only 0.01 percentage point. The exact-inertia controller

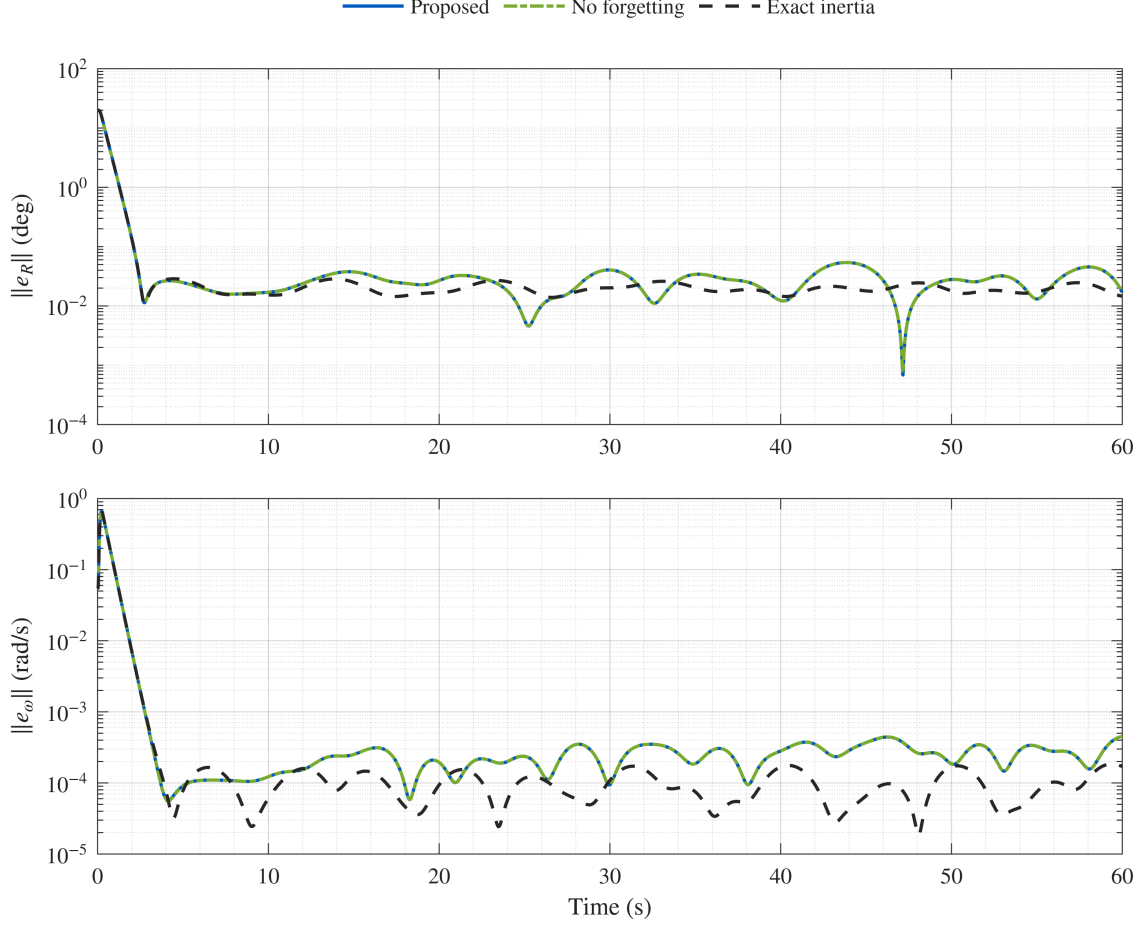


Fig. 6 Case 3 tracking errors under continuously varying full inertia, comparing the proposed, no-forgetting, and exact-inertia controllers using the common PCAC formulation.

has a slightly smaller attitude RMSE, whereas its full-interval angular-velocity RMSE is not smaller. These RMSE values include the common initial tracking transient.

The near-equivalence of the two adaptive controllers is consistent with weak activation of variable-rate forgetting. For the proposed controller,

$$\min_k \beta_k^{-1} \approx 0.982, \quad (111)$$

and β_k^{-1} remains close to one after initialization. The proposed estimator therefore behaves almost like ordinary RLS in this case. Under the selected settings, the slowly varying inertia produces only weak VRF activation. Consequently, Case 3 does not show a measurable VRF advantage. The comparison is nevertheless informative: together with the rapid-change result in Case 2, it indicates that the observed benefit of VRF depends on how strongly the parameter change activates forgetting. Weak VRF activation also produces no observable tracking penalty in this simulation.

Figure 7 shows that the proposed estimator follows the variation of the diagonal and off-diagonal inertia entries but does not track them exactly. Its mean relative inertia error is 7.43%. The maximum error of 43.12% occurs during

Table 3 Case 3 performance under continuously varying inertia.

Controller	Attitude RMSE (°)	Angular-rate RMSE (rad s ⁻¹)	Mean inertia error (%)	Maximum inertia error (%)	QP failures
Proposed	1.501	0.04814	7.43	43.12	0
No forgetting	1.501	0.04814	7.44	43.12	0
Exact inertia	1.499	0.04828	0	0	0

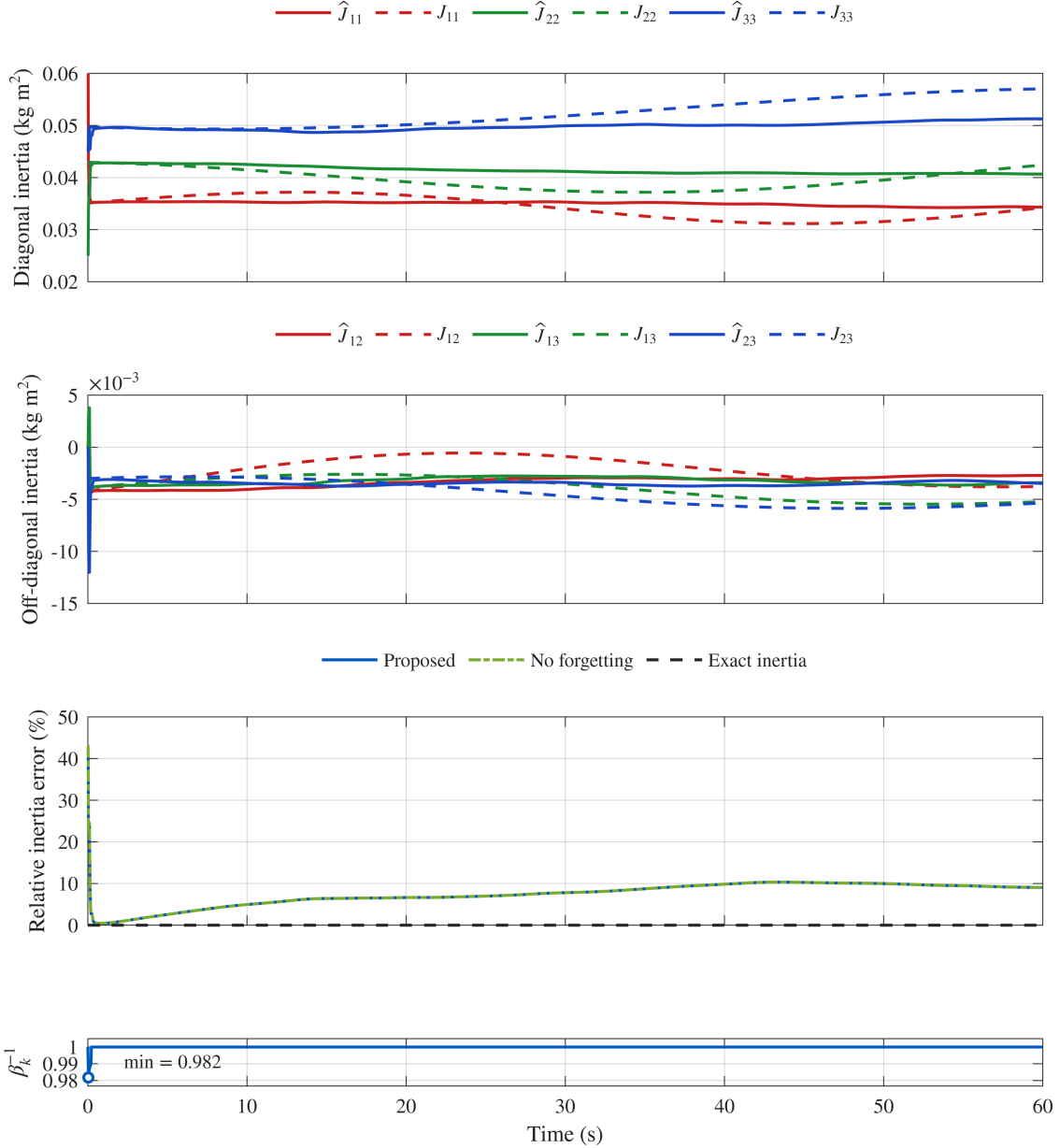


Fig. 7 Case 3 inertia estimates, relative identification errors, and instantaneous forgetting factor under continuous variation, showing the estimator's lag and residual bias.

484 initialization and therefore does not represent the sustained estimation error. After the initial correction, the estimate
485 retains visible lag and bias as the true inertia continues to vary.

486 This behavior is consistent with several interacting factors. The estimator must identify six coupled inertia parameters
487 using only the reference-induced and corrective closed-loop motion. No independent probing signal is applied, and
488 good regulation may reduce informative corrective motion. Moreover, the true inertia is a moving target rather than a
489 constant parameter. Because β_k^{-1} remains close to one, previously collected data retain substantial influence through the
490 RLS covariance recursion, which is consistent with the observed lag when the true inertia continues to vary. The large
491 initial covariance permits substantial initial correction but does not guarantee persistent excitation. Inertia projection
492 preserves physical admissibility but does not add information to the regression. Since measurement noise is zero, it
493 cannot explain the observed lag in this simulation. These points provide evidence-consistent explanations, but the
494 simulation does not isolate a unique cause.

495 Despite the remaining inertia error, the adaptive and exact-inertia controllers have similar full-interval tracking
496 metrics. Thus, this particular closed-loop trajectory is relatively insensitive to the observed model error. This result does
497 not imply complete identification or show that comparable inertia errors are harmless for other references, constraints, or
498 inertia profiles. Case 3 therefore demonstrates maintained closed-loop tracking under continuous full-inertia variation
499 while also showing the limitations of the selected estimator settings when tracking a slowly moving parameter without
500 dedicated excitation.

VIII. Conclusion

This paper developed a geometry-informed PCAC architecture for rigid-body attitude tracking with online full-inertia identification. The body frame need not coincide with the principal axes, and all six independent entries of the symmetric inertia matrix are estimated. An interval-integrated inverse-dynamics regression avoids numerical differentiation of measured angular velocity, while bounded positive-definite projection supplies a physically admissible model for prediction and control. The projected estimate is used in a nonlinear predictor that preserves the unit-quaternion structure and is expressed in local principal-rotation coordinates. Linearizing this predictor at the current operating point and freezing its Jacobian over the horizon produces an affine model. Consequently, torque and torque-increment constraints are enforced through a convex quadratic program.

The numerical studies delineate both the capabilities and limitations of this architecture. In the large-angle Monte Carlo study, the proposed controller stabilizes all 200 trials and identifies the full inertia substantially more accurately than the diagonal-adaptive and fixed-inertia controllers. Its tracking distributions remain close to those of the exact-inertia benchmark. This similarity is specific to the sampled conditions and does not imply that comparable model errors are harmless for other references, constraints, or inertia profiles. Following an abrupt full-inertia change, variable-rate forgetting yields near-complete reidentification and final tracking close to the exact-inertia benchmark, whereas the no-forgetting estimate remains biased. Under continuous inertia variation, the proposed and no-forgetting controllers have tracking metrics that agree to the reported precision. Forgetting is only weakly activated in this case and provides no measurable advantage, while both adaptive estimates retain lag and residual bias.

These results support the separation adopted in the framework: inverse dynamics are used for identification, nonlinear forward dynamics for prediction, and local affine dynamics for optimization. They also reinforce that accurate tracking and accurate parameter recovery are distinct objectives. Variable-rate forgetting can improve readaptation when a parameter change sufficiently activates residual-triggered forgetting, but it neither generates excitation nor guarantees convergence of all inertia parameters. The present evidence is limited to noise-free numerical cases and does not establish performance under measurement noise, unmodeled actuator dynamics, or hardware uncertainty. Future work will therefore consider experimental validation, excitation-aware adaptation, and extensions to more general time-varying rigid-body parameters.

Funding Sources

This research was partially supported by the MEXT SPReAD AI for Science program 2026 (Grant No. 26276994).

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